

How would you design a SmartDialer that retains the deterministic safety features of progressive dialing while maximizing the utilization advantage of predictive dialing?

The fundamental principle is to keep acting and decision-making distinct and never allow one element to perform both. The risk associated with predictive dialing is not the estimation itself, but rather the potential for an incorrect estimate to result in an abandoned, agent-less connected call. Therefore, a design that prevents a poor forecast from ever reaching the phone line unchecked is the solution rather than a more intelligent model.

The Predictive Pacing Engine is only meant to be used as a guide. With every tick, it calculates:

$$p_t = (1-\alpha) \cdot p_{(t-1)} + \alpha \cdot \text{outcome}_t \text{ (rolling answer rate)}$$
$$N_{\text{request}} \text{ (requested dial count)} = \text{floor} (h \cdot (1 - b) \cdot (A - R \cdot p) / p)$$

where h is a provider-health multiplier, b is a fixed safety buffer, R is calls that are currently ringing, and A is available agents. The reasoning behind $N < (A - R \cdot p) / p$ is simply capacity accounting: new calls shouldn't push the total expected connects past the available agents because calls that are already in flight will also convert at rate p . Because p is an estimate rather than a fact, h and b exist, making the request cautious in proportion to our current level of confidence in the data. Importantly, this engine just provides a number and an explanation; it makes no mention of the Call Allocator or the Telecom Provider. It's not a rule that someone might fail to follow; rather, it's a structural fact about the codebase.

Acting: Only the Safety Controller has the power to initiate a legitimate call. Because conditions can change in the seconds between a request and its execution, it rechecks ground truth (current A , current provider health) at the moment of execution rather than relying on the pacing engine's snapshot. When a guardrail is breached, it can either approve the request, lower it, reject it, or collapse the system to progressive behavior— $N = A$, one call per available agent—if there is insufficient data to trust p , provider health is below its floor, or the request has become stale in relation to the number of live agents.